PHASE-WISE ROADMAP (APRIL → JUNE/JULY) — WITH AI EDGE

PHASE 1: FOUNDATION (Week 1–2)

Goal:

Build Python basics + logical thinking + basic AI awareness

What you learn:

- Python:
  - Syntax, variables, loops, functions
  - Lists, dictionaries
- Problem-solving:
  - Very basic problems (patterns, math logic)

AI (LIGHT – 10–15 min/day only):

- What is AI vs ML
- Real-world use cases
- What is dataset, model (basic idea)

Output:

- Write basic programs independently
- Solve 40 problems
- Understand what AI is (no coding yet)

PHASE 2: CORE DSA START (Week 3–5)

Goal:

Build problem-solving muscle + introduce data thinking (AI base)

Topics:

- Arrays
- Strings
- Hashing
- Sliding window
- Two pointers
- Basic recursion

AI (CONTROLLED ADDITION):

- Install and use basics:
  - NumPy

    - Pandas (very basic)
- Learn:
    - What is dataset
    - Data cleaning basics

Total AI time: **3–4 hours across 3 weeks**

Output:

- Solve 100–120 problems
- Start recognizing patterns
- Understand how data is used in AI

PHASE 3: INTERMEDIATE DSA + CORE CS (Week 6–8)

Goal:

Become interview-capable + build first AI project

DSA:

- Linked List
- Stack & Queue
- Binary Search
- Trees (basic traversal)

Core Subjects (HIGH-YIELD ONLY):

- OS → Process, Threads, Scheduling
- DBMS → Joins, Indexing, Normalization
- OOP → Encapsulation, Inheritance, Polymorphism
- CN → HTTP, TCP/IP basics

AI (IMPORTANT START):

Week 6–7:

- Learn:
    - Supervised learning (concept)
    - Linear Regression (basic intuition)

Week 8:

- Build:
  Mini ML Project: Student Score Predictor

Tools:

- Python
- Basic ML library (no deep dive)

Output:

- 150–180 problems total
- Core CS basics clear
- 1 ML project built
- Can explain how AI works (basic level)

PHASE 4: ADVANCED + PROJECTS + INTERVIEW (Week 9–11)

Goal:

Placement-ready + AI differentiation

DSA:

- Graphs (BFS/DFS)
- DP (only 4–5 patterns)

PROJECTS (MODIFIED — HIGH IMPACT)

PROJECT 1 (MAIN):

AI-Powered Calories Tracker

Upgrade your existing project:

Features:

- User login + dashboard
- Calorie tracking
- AI diet suggestions
- Chatbot using:
  - OpenAI API
  - OR Google Gemini API

PROJECT 2:

Smart CRUD System

Example: Student Management

Add:

- Performance prediction (basic ML)
- Suggestions

PROJECT 3:

API-Based App

Example:

- Weather / News App

OPTIONAL PROJECT 4:

AI Chatbot

- Resume assistant / Study helper

SYSTEM DESIGN (BASIC ONLY):

- Client-server
- APIs
- Database design
- How AI API integrates

Output:

- 2–3 strong projects
- At least 1 AI-powered
- GitHub + Resume ready
- Confident explanation

PHASE 5: FINAL PUSH (Week 12 → DEADLINE)

Goal:

Crack interviews

Focus:

- Daily mock interviews
- DSA revision (patterns only)
- HR questions
- Aggressive applications

AI ADDITION:

Prepare answers:

- "How did you use AI?"
- "What problem does it solve?"
- "Why this approach?"

Output:

- Interview-ready
- Confident communication
- Strong differentiation

WEEKLY BREAKDOWN (UPDATED WITH AI)

Week	Focus	DSA Target	AI Task	MUST COMPLETE BY SUNDAY
Week 1	Python basics	20	AI concepts	Syntax + basics clear
Week 2	Python + logic	40	—	Functions + coding
Week 3	Arrays	60	Pandas intro	Array basics
Week 4	Strings + Hashing	80	Dataset basics	Pattern solving
Week 5	Recursion	100+	Data understanding	Recursion clarity
Week 6	Linked List	120	ML concept start	LL mastery
Week 7	Stack + Queue	140	Regression concept	Stack/Queue
Week 8	Trees + Core CS	160	ML project complete	Tree + AI project
Week 9	Graphs	170	AI project integration	Graph basics
Week 10	DP	190	AI features complete	2 projects done
Week 11	Revision	200+	AI explanation prep	Resume + mocks
Week 12	Final prep	Revise	—	Apply + interviews

DAILY SCHEDULE (UPDATED)

Time Block	Task
2.5–3 hrs	DSA (core focus)
1.5–2 hrs	Python / Core CS
1–1.5 hrs	Projects / AI
30–45 min	Revision

RESOURCES (UNCHANGED + AI ADD)

Area	Resource
Python	CodeWithHarry
DSA	LeetCode + Striver Sheet
Core CS	Gate Smashers
AI Basics	Krish Naik (selected videos)
AI APIs	Official docs (OpenAI / Google)

MOCK INTERVIEW STRATEGY

Stage	Plan
Week 10	1 mock every 2 days
Week 11–12	1 mock daily

Focus:

- Explaining logic
- Explaining projects (VERY IMPORTANT now)

COMMON MISTAKES (UPDATED)

Mistake	Impact
Ignoring DSA for AI	No placement
Too much AI theory	Time waste
Weak project explanation	Rejection
No mocks	Panic in interview

RECOVERY PLAN (UPDATED)

Problem	Solution
Behind schedule	Skip advanced AI
Less time	Focus only 1 AI project
Low confidence	Increase mocks
Overload	Drop optional project

FINAL CHECKLIST (UPDATED)

You are READY if:

- 180–220 DSA problems
- Solve medium problems
- 2–3 projects (1 AI-powered)
- Resume polished
- Core CS basics clear
- 8–10 mock interviews
- Applied to 50–100 companies